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Course Name – Data Analytics With GenAI

Subject – Excel

Topic – Logical Functions & Lookup

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Question 1:

Write a formula to check if the marks in cell B2 are greater than or equal to 40. If yes, return "Pass", otherwise "Fail".

Student Name	Marks
Riya	45
Siya	72
Raju	55
Rahul	80
Ram	30

Answer 1:

Formula in cell C2:

=IF($B2>=40,"Pass","Fail")

Explanation:

- IF checks if marks in B2 are greater than or equal to 40
- If YES → show "Pass"
- If NO → show "Fail"

Results:

- Riya (45) → **Pass**
- Siya (72) → **Pass**
- Raju (55) → **Pass**
- Rahul (80) → **Pass**
- Ram (30) → **Fail**

Drag the formula down to apply it to all rows.

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Question 2:

A company gives a bonus if sales are above 40,000 and attendance is greater than or equal to 90%. Write a formula to calculate whether the employee will get a bonus or not.

Employee	Sale	Attendance
Arjun	55000	95
Simran	65000	85
Rohit	30000	90
Anjali	80000	65

Answer 2:

Conditions:

- Sales > 40,000 **AND** Attendance ≥ 90% → Bonus
- Otherwise → No Bonus

Formula in cell D2:

=IF(AND($B2>40000,$C2>=90),"Eligible for Bonus", "Ineligible for Bonus")

Explanation:

- AND checks BOTH conditions must be true
- B2>40000 → Sales above 40,000
- C2>=90 → Attendance 90% or more

Results:

- Arjun: Sales=55000, Attendance=95 → **Eligible for Bonus**
- Simran: Sales=65000, Attendance=85 → **Ineligible for Bonus** (attendance < 90)
- Rohit: Sales=30000, Attendance=90 → **Ineligible for Bonus** (sales < 40000)
- Anjali: Sales=25000, Attendance=65 → **Ineligible for Bonus** (both conditions fail)

Drag the formula down to apply it to all rows.

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Question 3:

Write a nested IF formula:
If marks in B2 are ≥ 90 → "A", ≥ 75 → "B", ≥ 50 → "C", otherwise "Fail".

Student Name	Marks
Raj	85
Neha	96
Vivek	57
Rani	92
Sneha	75

Answer 3:

Conditions:

- If marks ≥ 90 → "A"
- If marks ≥ 75 → "B"
- If marks ≥ 50 → "C"
- Otherwise → "Fail"

Formula in cell C2:

=IF($B2>=90,"A", IF($B2>=75,"B",IF($B2>=50,"C","FAIL")))

Explanation:

- First checks if ≥90, if yes → "A"
- If not, checks if ≥75, if yes → "B"
- If not, checks if ≥50, if yes → "C"
- If none match → "Fail"

Results:

- Raj (85) → **B**
- Neha (96) → **A**
- Vivek (57) → **C**
- Roni (92) → **A**
- Sneha (75) → **B**

Drag the formula down to apply it to all rows.

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Question 4:

Find the Department of Employee ID = E103. Also explain which LOOKUP function is used and why?

EmpID	Name	Department
E101	Raju	IT
E102	Riya	HR
E103	Roshni	Finance
E104	Sejal	Sales

Answer 4:

Formula:
=VLOOKUP ("E103", $A2: $C5, 3, FALSE)
Answer: Finance

Which LOOKUP function is used and WHY?

VLOOKUP is used because:

1. **What is VLOOKUP?**
   - **V** stands for "Vertical"
   - It searches for a value in the **first column** and returns a value from another column in the same row
2. **Why VLOOKUP here?**
   - Employee IDs are in the **first column** (Column A)
   - We want to find the Department (Column C)
   - Data is arranged vertically in a table
3. **Formula breakdown:**
   - "E103" → What we're searching for
   - $A2:$C5 → Table range (all data)
   - 3 → Return value from 3rd column (Department)
   - FALSE → Exact match

Example explanation:

- Look for "E103" in column A
- When found (row 4), return the value from column C
- Result: **Finance**

Alternative using XLOOKUP (if available in newer Excel):
=XLOOKUP ("E103", A2:A5, C2:C5)

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Question 5:

A company has set the following commission structure for its sales team:

Sales Amount	Commission Rate
0	5%
20000	10%
50000	15%
80000	20%

If a salesperson achieves sales of ₹45,000, use the VLOOKUP function to determine the commission rate applicable.

Answer 5:

Formula:

=VLOOKUP(45000,A1:B5,2,TRUE)

Answer: 10%

Explanation:

- VLOOKUP searches for 45000 in the Sales Amount column
- TRUE means "approximate match" - it finds the closest value that's less than or equal to 45000
- 45000 falls between 20000 and 50000, so it uses the rate for 20000 → **10%**

To calculate actual commission amount:

=45000 * =VLOOKUP (45000,A1:B5,2,TRUE)

=45000 * 0.10

=₹4,500

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Question 6:

The HR department maintains the following leave records of employees: onus or not .

Employee Name	Leave Taken	Leave Allowed
Alex	10	12
Bob	10	15
Anna	8	12

Write an IF function to check whether an employee has Exceeded their allowed leave balance.

- **If Leaves Taken > Allowed Leaves, display "Exceeded".**
- **Otherwise, display "OK".**

Answer 6:

Formula in cell D2:

=IF($B2>$C2,"Exceeded","OK")

Explanation:

- Compares Leaves Taken (B2) with Leaves Allowed (C2)
- If Taken > Allowed → "Exceeded"
- Otherwise → "OK"

Results:

- Alex: 10 > 12? NO → **OK**
- Bob: 10 > 15? NO → **OK**
- Anna: 8 > 12? NO → **OK**

Example if someone exceeded:

- If Alex took 15 leaves but allowed only 12 → **Exceeded**

Drag the formula down to apply it to all rows.

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Question 7:

Find the salary of EmpID = E105 using INDEX-MATCH Function.

EmpID	Salary
E104	55000
E105	72000
E106	62000
E102	41000

Answer 7:

Formula:

=INDEX (B2:B5,MATCH("E105",A2:A5,0))

Answer: 72000

Explanation:

- **MATCH("E105", A2:A5, 0)** → Finds the position of E105 in column A
  - E105 is in the 2nd position
- **INDEX(B2:B5, 2)** → Returns the 2nd value from salary column
  - Result: **72000**

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Question 8:

Use the XLOOKUP function to find the Department of the employee with Emp ID = E303.

EmpID	Department
E101	HR
E303	IT
E106	Sales
E301	Finance

Answer 8:

Formula:

=XLOOKUP("E303",A2:A5,B2:B5,"Not Available",0,1)

Answer: IT

Explanation:

- Searches for "E303" in column A
- Returns corresponding value from column B (Department)
- Result: **IT**

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